

# Longitudinal Data Analysis: Chocolate cake data set analysis

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# Outline

- 1 The chocolate cake data set
- 2 Exploratory data analysis
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## Description of the data set

*An experiment was conducted to determine the effect of baking temperature on quality for three recipes of chocolate cake. Recipes I and II differed in that the chocolate was added at 40°C and 60°C, respectively, while recipe III contained extra sugar. Six different baking temperatures were considered: 175°C, 185°C, 195°C, 205°C, 215°C, and 225°C. 45 batches of cake batter were prepared using each of the 3 recipes 15 times in a completely random order. Each batch was large enough for 6 cakes, and the six baking temperatures were randomly assigned to the 6 cakes per batch in a random manner. The humidity, if the mold was greased (`Greased_mold`), and the amount of sunflower oil and palm oil (`Sunfloweroil_amount`, `Palmoil_amount`) in the oven was registered. One of several measurements of quality made on each cake was the breaking angle of the cake. The goal is to characterize these `BreakingAngles`".*

## Variables

- 270 observations, 8 different variables

| Variable            | Type        | Values                  |
|---------------------|-------------|-------------------------|
| Recipe              | Categorical | 1, 2, 3                 |
| Repetition          | Categorical | 1 to 15                 |
| Temperature         | Ordinal     | 175, 185, 200, 215, 225 |
| Sunfloweroil_amount | Ordinal     | 65, 150, 165            |
| Palmoil_amount      | Binary      | 0, 75                   |
| Greased_mold        | Binary      | False, True             |
| Humidity            | Continuous  | 1.109 - 17.708          |
| BreakingAngles      | Continuous  | 18 - 63                 |

**Table:** Description of the variables in our data set.

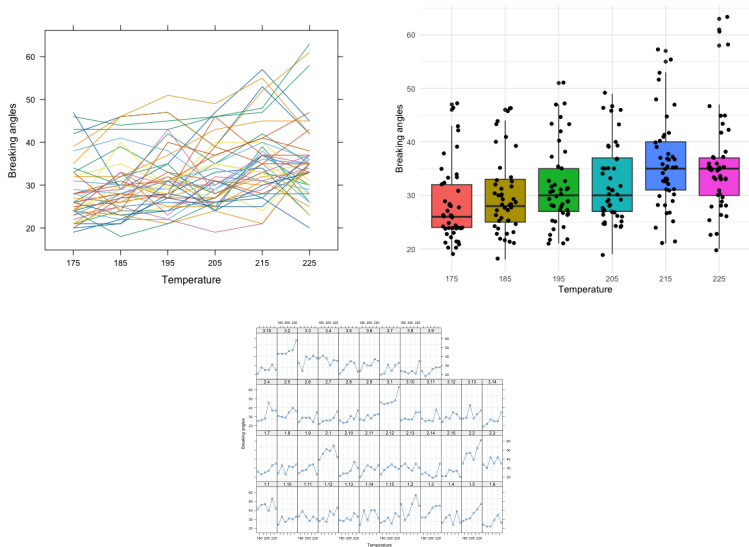
## Summary statistics

| <b>Outcome</b> | <b>Temp.</b> | <b>n</b> | <b>min</b> | <b>max</b> | <b>mean</b> | <b>sd</b> |
|----------------|--------------|----------|------------|------------|-------------|-----------|
| BrAngles       | 175          | 45       | 19         | 47         | 28.0        | 7.05      |
| BrAngles       | 185          | 45       | 18         | 46         | 30.0        | 7.26      |
| BrAngles       | 195          | 45       | 21         | 51         | 31.4        | 7.50      |
| BrAngles       | 205          | 45       | 19         | 49         | 32.2        | 7.35      |
| BrAngles       | 215          | 45       | 21         | 57         | 35.8        | 8.39      |
| BrAngles       | 225          | 45       | 20         | 63         | 35.4        | 9.03      |

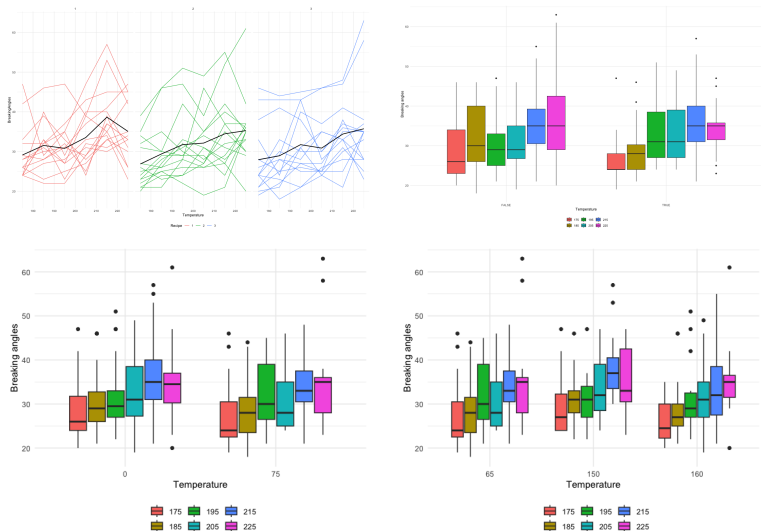
**Table:** Summary statistics for `BreakingAngles` according to `Temperature`.

## Summary statistics

- Total increment
- Increment per degree



**Figure:** Spaghetti plots per individual and boxplot of average evolution.



**Figure:** Mean evolution of the breaking angles across temperature for each subgroup of our variables.

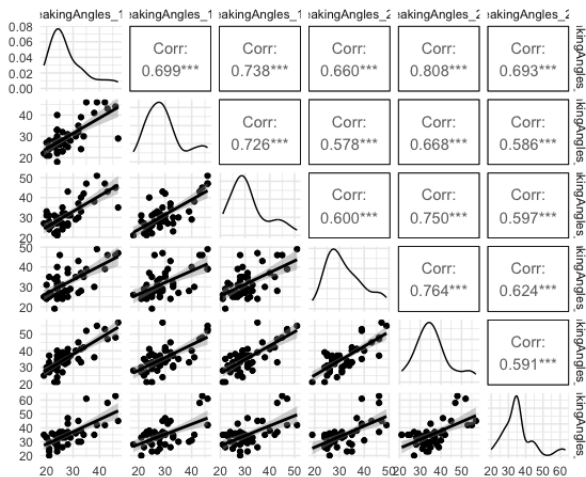


Figure: Correlation among measures.

## Conclusions of the exploratory analysis

- Our dataset has a continuous outcome, six categorical variables and a continuous one.
- Overall, the mean and variance of the breaking angles increase linearly with the temperature.
- The increment in the mean breaking angles is of 7.4 from the initial temperature to the final one, with an increment per unit of temperature of 0.15 degrees.
- It is hard to see differences among the subgroups of all the variables. Maybe between recipes there is a slight difference.
- There is variability between individuals that we might want to account for in future models.
- There is a high correlation between the measurements, and the first impression is that the correlation does not follow a particular structure.

## What is the interest of fitting a linear model?

- Get some first results and impressions knowing that this is not an adapt model
- Detect which covariates are the more relevant. We choose to use a stepwise selection, with the AIC criterion. By fitting this model, we get a first impression on the relevance of the data.
- Help the good development of more complicated model. Covariance between the variable. A first study of the mean structures.
- Detect the first traps of the data. A big variance, in the repetition of the data.

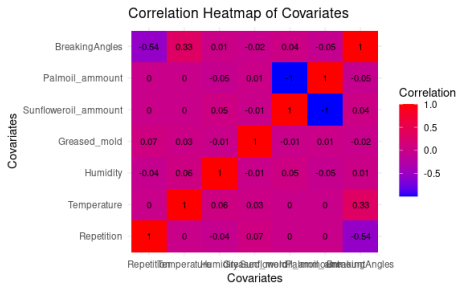
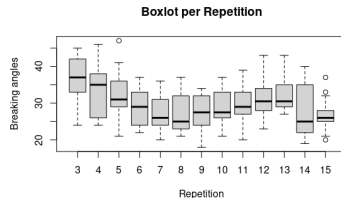
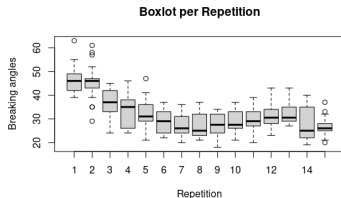


Figure: Correlation among covariates.



**Figure:** Boxplot of each repetition with and without the first 2 repetitions

## Result

- From this first try of fitting a model, we can conclude that the most relevant variables are: Repetition, Temperature.
- Variables Recipe and Greased\_mold are significant but less important compared to the two mentioned before.
- We can make the hypothesis that the variables Sunfloweroil\_amount, Palmoil\_amount, Humidity are not necessary.

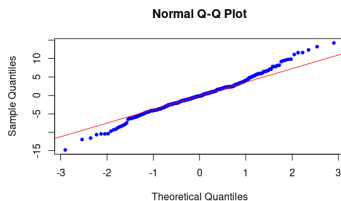
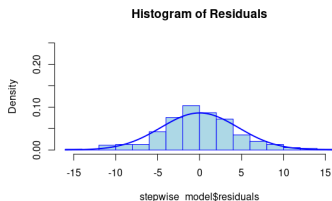


Figure: Checking some residuals

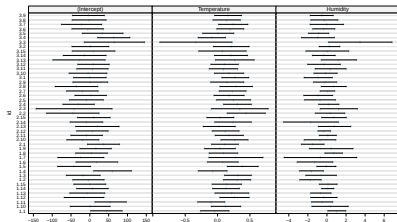
## First stage

The goal in this first stage is to explain the **within-subjects variability**. Let  $Y_{ij}$  denote our response of interest (breaking angle), for the  $i$ th individual (batch of cake), measured at temperature (remember that temperature plays the role of time in our problem)  $T_{ij}$ ,  $i = 1, \dots, N$ ,  $j = 1, \dots, n_i$ , that is,  $\mathbf{Y}_i = (Y_{i1}, \dots, Y_{in_i})'$ . In this first stage we assume that  $\mathbf{Y}_i$  satisfies the linear regression model

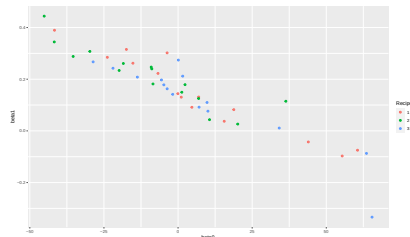
$$Y_{ij} = \beta_{1i} + \beta_{2i}T_{ij} + \beta_{3i}H_{ij} + \epsilon_{ij}, \quad j = 1, \dots, n_i. \quad (1)$$

Where  $\beta_{1i}$ ,  $\beta_{2i}$ ,  $\beta_{3i}$  are the unknown subject-specific regression coefficients.  $T_{ij}$  and  $H_{ij}$  are, respectively, the temperature and the humidity, in the  $j$ th measurement for the  $i$ th batch of cake. Finally,  $\epsilon_{ij}$  is the the error term.

# First stage



(a) Confidence intervals



(b)  $\beta_{1i}$  VS  $\beta_{2i}$

**Figure:** In the left, the confidence intervals of the estimates. In the right,  $\beta_{1i}$  VS  $\beta_{2i}$ .

The correlation between  $\beta_{1i}$  and  $\beta_{2i}$  is  $-0.93$ . The correlations between  $\beta_{1i}, \beta_{3i}$  and  $\beta_{2i}, \beta_{3i}$  are  $-0.023$  and  $-0.26$ , respectively.

## Two stage

The main objective in the second stage analysis is to explain the **variability between subjects**. To do this, we present the multivariate regression model

$$\left\{ \begin{array}{l} \beta_{1i} = \beta_1 R_{1i} + \beta_2 R_{2i} + \beta_3 R_{3i} + \sum_{l=1}^{15} \beta_{l+3} Rep_{li} + b_{1i} \\ \beta_{2i} = \beta_{19} R_{1i} + \beta_{20} R_{2i} + \beta_{21} R_{3i} + \sum_{l=1}^{15} \beta_{l+21} Rep_{li} + b_{2i} \\ \beta_{3i} = \beta_{39} R_{1i} + \beta_{40} R_{2i} + \beta_{41} R_{3i} + \sum_{l=1}^{15} \beta_{l+41} Rep_{li} + b_{3i} \end{array} \right. \quad (2)$$

Where  $R_{1i}, R_{2i}, R_{3i}$  are indicator variables defined to be one if the batch  $i$  is from Recipe 1, 2 or 3 respectively. And  $Rep_{1i}, \dots, Rep_{15i}$  are also indicator variables which are one if batch  $i$  belongs to the repetition 1,  $\dots$  or 15.

## Two stage

- $\beta_1, \beta_2, \beta_3$  represents the average response at the beginning (first temperature used) for each recipe. Same for  $\beta_4$  to  $\beta_{18}$ , this time for each repetition.
- The parameters  $\beta_{19}, \beta_{20}$  and  $\beta_{21}$  represents the average temperature effects for each recipe separately. Moreover, the parameters  $\beta_{22}$  to  $\beta_{38}$  represents the average temperature effects for each repetition separately.
- Analogously we can interpret the meaning of the remaining parameters, these will be related with humidity.

## Results

| Coefficient  | Estimate | Std Error | T Value | P Value |
|--------------|----------|-----------|---------|---------|
| (Intercept)  | 31.78    | 16.56     | 1.92    | 0.07    |
| Recipe2      | -15.82   | 9.84      | -1.61   | 0.12    |
| Recipe3      | 0.89     | 9.84      | 0.09    | 0.93    |
| Repetition2  | -42.55   | 22.00     | -1.93   | 0.06    |
| Repetition3  | -24.71   | 22.00     | -1.12   | 0.27    |
| Repetition4  | 4.65     | 22.00     | 0.21    | 0.83    |
| Repetition5  | -32.34   | 22.00     | -1.47   | 0.15    |
| Repetition6  | -21.61   | 22.00     | -0.98   | 0.33    |
| Repetition7  | -48.69   | 22.00     | -2.21   | 0.04    |
| Repetition8  | -37.64   | 22.00     | -1.71   | 0.10    |
| Repetition9  | -22.41   | 22.00     | -1.02   | 0.32    |
| Repetition10 | -34.19   | 22.00     | -1.55   | 0.13    |
| Repetition11 | -8.79    | 22.00     | -0.40   | 0.69    |
| Repetition12 | -26.30   | 22.00     | -1.20   | 0.24    |
| Repetition13 | -31.89   | 22.00     | -1.45   | 0.16    |
| Repetition14 | -32.00   | 22.00     | -1.45   | 0.16    |
| Repetition15 | -19.92   | 22.00     | -0.91   | 0.37    |

**Table:** Summary of the linear model fitted for the first stage  $\beta_{1i}$ .

| Coefficient  | Estimate | Std Error | T Value | P Value |
|--------------|----------|-----------|---------|---------|
| (Intercept)  | 0.10     | 0.09      | 1.12    | 0.27    |
| Recipe2      | 0.07     | 0.05      | 1.24    | 0.23    |
| Recipe3      | -0.03    | 0.05      | -0.52   | 0.61    |
| Repetition2  | 0.20     | 0.12      | 1.68    | 0.10    |
| Repetition3  | -0.01    | 0.12      | -0.06   | 0.96    |
| Repetition4  | -0.07    | 0.12      | -0.55   | 0.59    |
| Repetition5  | 0.10     | 0.12      | 0.83    | 0.42    |
| Repetition6  | 0.04     | 0.12      | 0.31    | 0.76    |
| Repetition7  | 0.14     | 0.12      | 1.14    | 0.26    |
| Repetition8  | 0.06     | 0.12      | 0.48    | 0.63    |
| Repetition9  | 0.00     | 0.12      | 0.01    | 0.99    |
| Repetition10 | 0.07     | 0.12      | 0.61    | 0.54    |
| Repetition11 | -0.06    | 0.12      | -0.47   | 0.65    |
| Repetition12 | 0.05     | 0.12      | 0.41    | 0.68    |
| Repetition13 | 0.06     | 0.12      | 0.47    | 0.64    |
| Repetition14 | 0.08     | 0.12      | 0.64    | 0.53    |
| Repetition15 | -0.03    | 0.12      | -0.23   | 0.82    |

**Table:** Summary of the linear model fitted for the first stage  $\beta_{2i}$ .

| Coefficient  | Estimate | Std Error | T Value | P Value |
|--------------|----------|-----------|---------|---------|
| (Intercept)  | -0.64    | 0.55      | -1.16   | 0.25    |
| Recipe2      | 0.19     | 0.33      | 0.58    | 0.57    |
| Recipe3      | 0.37     | 0.33      | 1.12    | 0.27    |
| Repetition2  | 0.08     | 0.73      | 0.11    | 0.91    |
| Repetition3  | 1.83     | 0.73      | 2.50    | 0.02    |
| Repetition4  | -0.33    | 0.73      | -0.45   | 0.65    |
| Repetition5  | -0.09    | 0.73      | -0.13   | 0.90    |
| Repetition6  | -0.38    | 0.73      | -0.51   | 0.61    |
| Repetition7  | 0.24     | 0.73      | 0.33    | 0.74    |
| Repetition8  | 0.84     | 0.73      | 1.15    | 0.26    |
| Repetition9  | 0.36     | 0.73      | 0.49    | 0.63    |
| Repetition10 | 0.28     | 0.73      | 0.38    | 0.71    |
| Repetition11 | 0.45     | 0.73      | 0.61    | 0.55    |
| Repetition12 | 0.16     | 0.73      | 0.21    | 0.83    |
| Repetition13 | 0.93     | 0.73      | 1.27    | 0.21    |
| Repetition14 | -0.18    | 0.73      | -0.25   | 0.81    |
| Repetition15 | 0.76     | 0.73      | 1.04    | 0.31    |

**Table:** Summary of the linear model fitted for the first stage  $\beta_{3i}$ .

Few estimates are significant at a 0.05 level. All the estimates for the parameter  $\beta_{2i}$  (see Table 4), which is the one related with the temperature, are all near to 0, and non-significant.

## Combined model

There is the combined model:

$$\begin{aligned}
 Y_{ij} = & (\beta_1 R_{1i} + \beta_2 R_{2i} + \beta_3 R_{3i} + \sum_{l=1}^{15} \beta_{l+3} \text{Repl}_{li} + b_{1i}) + \\
 & (\beta_{19} R_{1i} + \beta_{20} R_{2i} + \beta_{21} R_{3i} + \sum_{l=1}^{15} \beta_{l+21} \text{Repl}_{li} + b_{2i}) T_{ij} + \\
 & (\beta_{39} R_{1i} + \beta_{40} R_{2i} + \beta_{41} R_{3i} + \sum_{l=1}^{15} \beta_{l+41} \text{Repl}_{li} + b_{3i}) H_{ij} + \epsilon_{ij}.
 \end{aligned} \tag{3}$$

## Random intercept

We have fitted a **Linear Mixed Model** (LMM) with a **random intercept**, that is

$$Y_{ij} = (\beta_0 + b_{0i}) + \beta_1 T_{ij} + \epsilon_{ij}, \quad (4)$$

where  $b_0 \sim N(0, d_{00})$  and  $\epsilon \sim N(0, \sigma^2)$ . It is important to remark that these  $b_0$ 's are now random effects, while in the two-stage analysis they were not.

Notice that the random intercept accounts for the variability explained by the Repetition covariate in the Temperature baseline.

## Results

| Random effects:               |           |            |          |
|-------------------------------|-----------|------------|----------|
| Groups                        | Name      | Variance   | Std.Dev. |
| id                            | Intercept | 40.25      | 6.34     |
| Residual                      |           | 20.71      | 4.55     |
| Fixed effects:                |           |            |          |
|                               | Estimate  | Std. Error | t value  |
| (Intercept)                   | 0.516     | 3.389      | 0.152    |
| Temperature                   | 0.158     | 0.016      | 9.746    |
| Correlation of Fixed Effects: |           |            |          |
|                               | (Intr)    |            |          |
| Temperature                   | -0.957    |            |          |

**Table:** Summary of the random and fixed effects of the model (4).

## Comparing models

We compare with the normal linear model

$$Y_{ij} = \beta_0 + \beta_1 T_{ij} + \epsilon_{ij}.$$

- $\lambda_{LR} = 170$  chi-squared distributed by the Wilks' theorem.
- $AIC = 1877.304$ ,  $BIC = 1888.099$  in the normal linear model.
- $AIC = 1709.298$ ,  $BIC = 1723.692$  in the random intercept model.

## Random slope

Now, we fit a **random slope** model, it is

$$Y_{ij} = \beta_0 + (\beta_1 + b_{1i})T_{ij} + \epsilon_{ij}, \quad (5)$$

where  $b_{1i} \sim N(0, d_{11})$ .

## Results

| Random effects:               |             |            |          |
|-------------------------------|-------------|------------|----------|
| Groups                        | Name        | Variance   | Std.Dev. |
| id                            | Temperature | 0.001      | 0.032    |
| Residual                      |             | 20.45      | 4.52     |
| Fixed effects:                |             |            |          |
|                               | Estimate    | Std. Error | t value  |
| (Intercept)                   | 0.516       | 3.235      | 0.159    |
| Temperature                   | 0.158       | 0.0168     | 9.409    |
| Correlation of Fixed Effects: |             |            |          |
|                               | (Intr)      |            |          |
| Temperature                   | -0.956      |            |          |

**Table:** Summary of the random and fixed effects of the model (5).

## Random intercept and random slope

Let us fit a random-effects model with a **random intercept and a random slope** and check the results. The model is the following,

$$Y_{ij} = (\beta_0 + b_{0i}) + (\beta_1 + b_{1i})T_{ij} + \epsilon_{ij}, \quad (6)$$

where  $(b_0, b_1) \sim N(0, D)$  and  $\epsilon_i \sim N(0, \Sigma_i)$ .

When fitting the previous model in R we have **convergence errors**. Using either restricted maximum likelihood (REML) criterion or maximum likelihood.

## Playing with variance-covariance structure

However, by taking simpler **variance-covariance** matrix of the random effects than the unstructured one, we are able to fit the model. We have fitted (6) taking

$$D_d = \begin{pmatrix} \sigma_1^2 & 0 \\ 0 & \sigma_2^2 \end{pmatrix} \quad \text{or} \quad D_{id} = \begin{pmatrix} \sigma_1^2 & 0 \\ 0 & \sigma_1^2 \end{pmatrix}.$$

- In terms of AIC and BIC the model using  $D_{id}$  is better.  
 $AIC = 1706.794, BIC = 1721.158$  in the case of  $D_{id}$ , and  
 $AIC = 1708.751, BIC = 1726.706$  in the  $D_d$  case.
- Moreover, the model above, taking  $D_{id}$ , is forcing the two random effects (intercept and slope) to have the same variance.

## Taking some transformation

We try to take the logarithm in the response variable, obtaining the following model,

$$\log(Y_{ij}) = (\beta_0 + b_{0i}) + \beta_1 T_{ij} + \epsilon_{ij}, \quad (7)$$

where  $b_0 \sim N(0, d_{00})$  and  $\epsilon \sim N(0, \sigma^2)$

## Results

| Random effects:               |           |            |          |
|-------------------------------|-----------|------------|----------|
| Groups                        | Name      | Variance   | Std.Dev. |
| id                            | Intercept | 0.0335     | 0.183    |
| Residual                      |           | 0.0194     | 0.139    |
| Fixed effects:                |           |            |          |
|                               | Estimate  | Std. Error | t value  |
| (Intercept)                   | 2.4503602 | 0.1035     | 23.734   |
| Temperature                   | 0.0049    | 0.0005     | 9.967    |
| Correlation of Fixed Effects: |           |            |          |
|                               | (Intr)    |            |          |
| Temperature                   | -0.961    |            |          |

**Table:** Summary of the random and fixed effects of the model (7).

## Comparing models

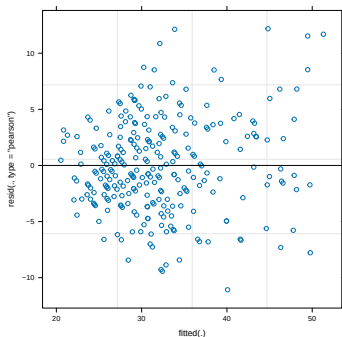
- **Much lower variance** than in the model without transformation (0.0335 and 40.25).
- **Much lower AIC and BIC**,  $AIC = -164.5144$ ,  $BIC = -150.121$  against  $AIC = 1709.298$ ,  $BIC = 1723.692$ .

In conclusion, the transformed model is the best one.

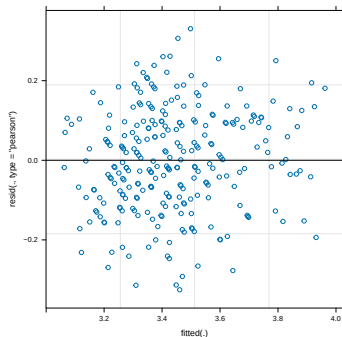
## Further analysis

- Several structures of **correlation matrix** tried: compound symmetry, auto-regressive and general correlation matrix with no structure. No one improves the model.
- The usage of **weights for model the variance** do not improve the model performance.

# Residuals



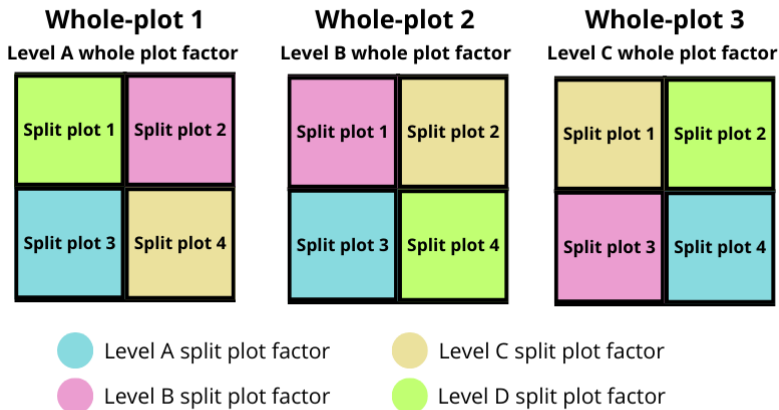
(a) Residuals of the random-effects model **without** transformation



(b) Residuals of the random-effects model **with** transformation

**Figure:** In the left, Pearson's residuals against the fitted values for the random effect model without taking the logarithm. In the right, the same but for model taking logarithm.

# Split-plot models



## Split-plot model for our data

- Whole plot  $\implies$  Batch
- Whole plot factor  $\implies$  Recipe
- Split plots  $\implies$  Cake
- Split plot Factor  $\implies$  Temperature

## Split-plot model for our data

| Batch 1        |               | Batch 2       |               | ... | Batch 15      |               |
|----------------|---------------|---------------|---------------|-----|---------------|---------------|
| Recipe 1       |               | Recipe 1      |               |     | Recipe 3      |               |
| Cake 1<br>T° 1 | Cake 2<br>T°2 | Cake 1<br>T°1 | Cake 2<br>T°2 |     | Cke 1<br>T°1  | Cake 2<br>T°2 |
| Cake 3<br>T°3  | Cake 4<br>T°4 | Cake 3<br>T°3 | Cake 4<br>T°4 |     | Cake 3<br>T°3 | Cake 4<br>T°4 |
| Cake 5<br>T°5  | Cake 6<br>T°6 | Cake 5<br>T°5 | Cake 6<br>T°6 |     | Cake 5<br>T°5 | Cake 6<br>T°6 |

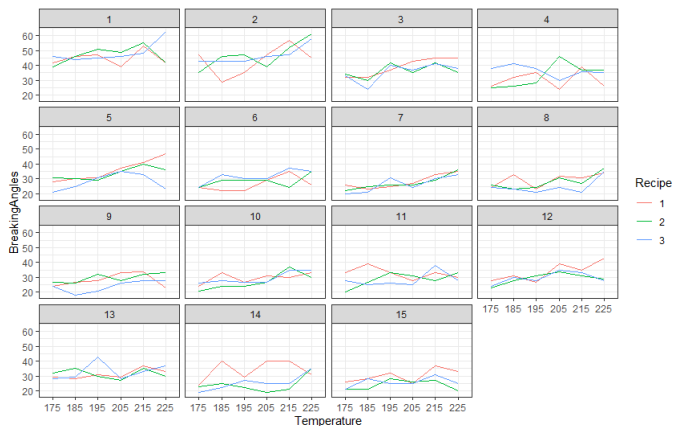
**Table:** Structure of our problem

$$Y_{ijk} = \mu + \alpha_i + \eta_{k(i)} + \beta_j + (\alpha\beta)_{ij} + \epsilon_{ijk},$$

## Results of the model

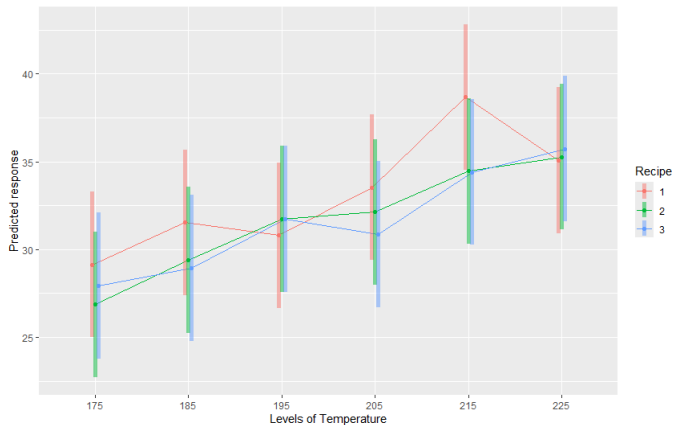
- The model is valid for our data.
- At level 0.05, the fixed effect of temperature is significant.
- Recipe is not a significant factor for the response (Breaking Angle).
- Interaction between recipe and temperature is not a significant factor for the response .

## Results of the model



**Figure:** Spaghetti plot grouped by recipe in each batch

## Model mean prediction



**Figure:** Predicted response for each temperature and recipe by our model,

See more code and material in our GitHub repositories:

- <https://github.com/adinbartoli/LDA>
- [https://github.com/ArnauGF/MSc-UPC/tree/longitudinal\\_data\\_analysis](https://github.com/ArnauGF/MSc-UPC/tree/longitudinal_data_analysis)